

Xingzhi Qian

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Education

Cornell University

New York, United States

Master of engineering in Electrical and Computer Engineering

Aug. 2025–Dec. 2026

Shanghai Normal University

Shanghai, China

B.Eng. in Electronic Information Engineering (Joint Program with University of Dayton)

Sept. 2021 – Jun. 2025

GPA: 3.81 /4.0 | Class Rank: 2/83 | College Rank: 2/512

Publication

- Hu, Yu, An, Kang, Liu, Xiangpeng, Ji, Yixuan, **Qian, Xingzhi**, et al. “Optimizing Path Planning and Conflict Resolution in High-Density AS/RS with a Multi-Shuttle Cooperative Scheduling Framework.” IEEE Transactions on Instrumentation & Measurement, Submitted.
- Z.-H. Cheng, Z. -Y. Lin, and **X. -Z. Qian, (Co-author)**. “Endoscopic-based 3D reconstruction methods of human internal cavity.” Proceedings of 2023 International Conference on Computer Science and Mechatronics, vol. 3017, no. 1, 2023

Research

During graduate studies:

Teaching Assistant in “ECE/ENGRD 2720. Data Science for Engineers” in Cornell University

- Served as a teaching assistant for a core data science course for engineering students. The course covered the full data science workflow, including data acquisition and cleaning, exploratory analysis, modeling and prediction, visualization, and presentation. It systematically taught core techniques such as numerical optimization, the Discrete Fourier Transform, Principal Component Analysis, statistical inference, regression and classification, outlier detection, and overfitting avoidance, while also emphasizing ethics in data driven decision making and the integration of data science with engineering contexts.
- Supported course instruction and student development, focusing on the mathematical foundations and practical implementation of data science methods. Assisted with Python programming projects, explained core algorithm concepts, helped students debug code and interpret results, and provided one on one guidance on applying data science techniques to real engineering problems. Helped students master the full process from algorithm design to data analysis and result presentation, successfully completing the teaching objectives.

Raspberry Pi based smart classroom assistant

Project link:<https://github.com/haowang854/Classroom-Assistant>

- Designed and implemented a distributed embedded control system for a smart classroom. Built a hardware architecture with a Raspberry Pi as the central controller and two ESP8266 boards as peripheral nodes. Developed functions for the main controller including touchscreen interaction, speech recognition, MQTT communication, and global state management, as well as sensor data acquisition and actuator control on the peripheral nodes.
- Set up a stable MQTT communication framework based on the Mosquitto protocol, enabling real time data exchange and command transmission between the master and slave devices. Integrated multiple sensors, accurately collecting temperature, humidity (DHT22), and illuminance (BH1750) data, and controlled actuators including a fan, heater, WS2812B LED strips, and a DC motor for a projection screen.
- Designed and optimized adaptive environmental control logic. Introduced hysteresis control for temperature and humidity regulation to reduce frequent device switching. Implemented adaptive dimming, multi mode lighting control, and priority based command overrides. Built three scenario based control modules: Projection Mode, Movie Mode, and Off Mode. Used state machine design and command interlocking to ensure stable and safe open loop motor control without encoders.
- Completed full system end to end integration, connecting touchscreen control, voice commands, sensor feedback, actuator driving, and global state management. Gained practical experience in distributed embedded architecture, multi device communication, open loop control optimization, and robust state machine design, delivering a functional smart classroom prototype.

Option Pricing and Hedging with Calibrated SDE Models

Advisor: Vikram Krishnamurthy

Completed on: Cornell Statistical Signal Processing Laboratory

- Built a stochastic volatility option pricing and dynamic hedging framework using real market data for SPY call options. Implemented and systematically evaluated the Neural Heston stochastic differential equation model against a geometric Brownian motion baseline model.
- Completed full quantitative modeling workflows, including Monte Carlo simulation for pricing, parameter calibration by minimizing pricing error against market quotes, robust pricing interval estimation through lower and upper bound optimization, and dynamic hedging using a neural hedge network. Covered the entire process of quantitative pricing and risk management.
- Achieved significant breakthroughs and practical outcomes. The Neural Heston model notably outperformed the GBM baseline in pricing accuracy, produced robust pricing intervals that fit actual market prices, and reduced hedging variance by 71.46%, demonstrating the core value of stochastic volatility modeling for improving pricing accuracy and financial risk management.

Cooperative Multi-Agent Learning in Overcooked-AI

- Built a cooperative Markov game model with two agents in the Overcooked AI simulation environment, focusing on multi agent collaborative decision making in shared scenarios. Investigated how reward function design affects agent coordination, learning stability, and task completion efficiency.
- Independently built a reinforcement learning pipeline using Python and PyTorch. Designed and compared multiple reward shaping schemes including sparse team rewards and shaped rewards for intermediate subtasks. Systematically evaluated algorithm performance and collaborative strategies using metrics such as total task return, number of completed dishes, training convergence curves, and qualitative analysis of policy behavior.
- Explored core issues in multi agent collaborative decision making, focusing on reward function optimization, agent coordination mechanisms, policy iteration and evaluation, and the emergence of cooperative behavior in sequential decision making environments. Formed a complete experimental framework and research conclusions for multi agent reinforcement learning.

Fast robot course design

Project link: <https://alanq0327.github.io/>

- The course focused on system level design and implementation of dynamic autonomous robots. The core objective was to develop a high speed autonomous car, exploring dynamic behavior control, force analysis, sensor integration, and reactive control techniques on an embedded platform.
- Responsible for system development based on the Artemis Nano main control board. Integrated hardware modules including an inertial measurement unit and a time of flight distance sensor. Designed a low latency software architecture with noise tolerant implementation, while also exploring the benefits of partial off board computation for system performance optimization.
- Built a complete embedded control pipeline, achieving sensor data acquisition, noise filtering, state estimation, and real time reactive control. Solved key engineering problems such as stability control, latency optimization, and noise tolerance in high speed motion scenarios.
- Used a peer to peer mentoring model, iteratively improving project solutions under instructor guidance, uploading development progress and design ideas, and contributing to a reusable project knowledge base to provide a practical framework for future courses and research.

During undergraduate studies

Face Recognition and Automatic Sign-In System for Complex Classroom Scenes

Advisor: Prof. Kang An

Research Member

Nov. 2023 – Nov. 2024

Project description: Collaboratively built a system that integrates a trained deep learning model (YOLOv8 and FaceNet) for face recognition and person counting in challenging classroom scenes. The system includes a variable zoom pan tilt camera, an automatic sign in program, and a management app.

- Improved the network structures of YOLOv8 and FaceNet by sharing convolutional layers, adding attention mechanisms, and using feature fusion to enable simultaneous face recognition and people counting.
- Collected classroom images with various obstacles (occluded faces, side angles) and environmental challenges (low resolution, uneven lighting, misaligned illumination). Tackled facial challenges by making the algorithm sensitive to pixel value edges and textures, and environmental issues using closed set data and 3D panorama stitching.
- Programmed the auto sign in program that connects to a live camera and a Bluetooth linked management app.
- Designed the mechanical structure of the variable focus pan tilt unit to eliminate blind spots in diverse classroom layouts.

YOLOv5 Algorithm Based Traffic Light Recognition Guide Cane with Voice Reminder

Advisor: Prof. Bo Zhang and Prof. Xiao Lin

Lead Researcher (Funded Project of Shanghai College Students Innovation & Entrepreneurial Training Program)

Jun. 2023 – Aug. 2024

- Trained a YOLOv5 model on traffic light images taken under various conditions, tuned parameters, and refined training strategies. Performed extensive field tests and continuously improved the model to ensure stable and reliable performance in real world settings.
- Used Google APIs and gTTS to code a voice interaction module that provides real time spoken feedback.

- Created detailed mechanical drawings for the cane and successfully integrated the machine vision and voice modules into the body of the cane.

Optimizing Path Planning and Conflict Resolution in High-Density AS/RS with a Multi-Shuttle Cooperative Scheduling Framework

Principal Investigator: Prof. Kang An

Research Member

Jul. 2023 – May 2024

Project description:Contributed to a project that proposed a grid map based Multi Shuttle Cooperative Scheduling Model (MSCSM) to tackle individual shuttle path planning and multi shuttle collision avoidance in automated storage and retrieval systems.

- Assisted the principal investigator in developing a Fuzzy A* path planning algorithm (FMOSAstar) that considers both turning cost and the side with fewer obstacles (the side of the straight line connecting start and end points). This algorithm balances efficiency with path length, reduces path conflicts in dense areas, and minimizes turning points.
- Helped design a grid map based adaptive time window algorithm (GMATW) to improve shuttle and track utilization and overcome problems caused by single avoidance rules, such as poor handling of unexpected tasks and resulting idle time.
- Validated the MSCSM through experiments in both real and simulated warehouse environments.

Achievement: This research led to a paper submitted to IEEE Transactions on Instrumentation & Measurement, with the author as a co author.

An OpenMV-Based Autonomous Navigation and Pesticide Fixed-Point Spray Robot

Advisor: Prof. Ye Yang

Lead Researcher （Advisor: Prof. Ye Yang）

Nov. 2022 – Nov. 2023

Project description:Led a team to develop a robot capable of autonomous navigation (using SLAM), movement, visual recognition of plant damage (OpenMV), and precise pesticide spraying on targeted areas with controlled spray volume and time (STM32).

- Designed and assembled the overall robot, including its mechanical structure, modules, and working mechanisms.
- Coordinated team members to complete sketches of the robot's mechanical structure and units.
- Programmed, adjusted, and implemented the robot's autonomous navigation and movement module and target recognition module. Participated in testing, debugging, and adjusting other modules' code.
- Researched and selected appropriate accessories for the robot.

Achievement: The built prototype won Second Prize in the 2023 National Undergraduate Mechanical Innovation Competition and a utility model patent.

Internships

Shanghai Longcheer Technology Co., Ltd. (603341: XSHG)

Shanghai, China

Hardware Test Intern

Aug.2024 – Dec. 2024

- Designed reasonable test plans for Longcheer's Universal Fast Charging Specification (UFCS) and dual cell battery fast charging products according to specifications, executed tests, produced reports, managed risks at each R&D milestone, and tracked issue resolution.
- Tested intelligent charging products under various scenarios with different temperatures (both fixed and changing), remaining battery levels, and app usage.
- Participated in signal performance testing of adapters to verify that communication frequency between the MCU and adapter, bit and byte intervals, and DP delay met requirements. Proposed an improvement that was adopted by the company.

Hangzhou Bailong Electronics Co., Ltd.

Hangzhou, China

Hardware Design Intern

Jul.2023 – Aug. 2023

- Simulated and analyzed engineering processes using LabVIEW, summarized results, and produced technical analysis documents.
- Integrated functional requirements of intelligent hardware, participated in hardware solution design and implementation, and drew schematics and PCB layouts.
- Completed chip performance testing and analysis, designed test plans and test cases, executed tests, and recorded results to support the team's subsequent research.

HONORS AND AWARDS

Second Prize in National Undergraduate Mechanical Innovation Competition, by National Undergraduate Mechanical Innovation Competition Committee

May 2023

First Prize in "Zhengda Cup" National College Student Market Research and Analysis Competition, by Commerce Statistical Society of China

Apr. 2023

Second Prize in Contemporary Undergraduate Mathematical Contest in Modeling, by China Society for Industrial and Applied Mathematics

Nov. 2022

First-class Academic Scholarship, Shanghai Normal University

2023/2022/2021

Shanghai Outstanding Graduate

Jun. 2025

PROFESSIONAL SKILLS

Languages: Mandarin, English (IELTS 7.0), Japanese (JLPT N1), Cantonese (business level)

Software Tools: Altium Designer, Multisim, Proteus, Keil, Vivado, PyCharm, Matlab, Arduino, LabVIEW, Solidworks, AutoCAD, Git, SPSS, Adobe Suite (AI, AN, DW, PS)

Programming: Python, C, C++, Matlab, Visual Basic, Verilog, JavaScript

Machine Learning Frameworks: Scikit Learn, TensorFlow, PyTorch

Machine Learning Algorithms: Convolutional Neural Networks (CNNs), XGBoost, LightGBM, Long Short Term Memory (LSTM), Logistic Regression, Principal Component Analysis (PCA), K Means Clustering, K Nearest Neighbors (KNN), YOLO